

Assignment 1: Student Performance Prediction Using ANN

Complete Project Guide aur Assignment Details (Roman Urdu Format)

1. OBJECTIVE (MAQSAD)

Aap ko **Student Performance Dataset** diya ja raha hai. Is assignment mein aap ne **Artificial Neural Network (ANN)** ka use karte hue students ki performance predict karni hai.

Dataset Overview:

- **Dataset Name:** 300K Student Performance Prediction Dataset
- **Dataset Link:**

<https://www.kaggle.com/datasets/rhythmghai/300k-student-performance-prediction-dataset>

2. TASK

Dataset ko properly preprocess karne ke baad ANN model develop karein aur given output/target column ki prediction karein.

- **Output Column (Target):** Final Grade
- Aap ke input features (X) mein target column ko include nahi karna. Sirf relevant input features ko ANN mein dena hai.

3. REQUIREMENTS

1. Data Preprocessing

- Dataset load karein.

- Dataset ka basic EDA (Exploratory Data Analysis) perform karein.
- Missing values check karein aur properly handle karein.
- Duplicate values check karein.
- Categorical data agar present ho to appropriate encoding perform karein.
- Features (X) aur target (y) separate karein.
- Dataset ko training aur testing data mein divide karein.
- Numerical features par Feature Scaling apply karein.

2. ANN Model Building

ANN model TensorFlow/Keras ka use kar ke develop karein. Aap ko different ANN architectures experiment karni hain, for example:

- Different number of hidden layers
- Different number of neurons
- Different activation functions
- Different optimizers
- Different epochs aur batch sizes

3. Activation Functions

Kam az kam 3 different activation functions/configurations experiment karein:

- ReLU
- Tanh
- Sigmoid

4. Optimizers

Kam az kam 3 different optimizers try karein:

- Adam
- SGD
- RMSprop

5. Model Training & Monitoring

Har model ko train karein aur training ke dauran Loss, Validation Loss, aur Performance Metrics observe karein. Different epochs try karein (for example: 20, 50, 100).

6. Prediction

Best performing ANN model select karne ke baad Test data par prediction perform karein. Kuch sample predictions display karein aur Actual value aur predicted value compare karein.

7. Model Comparison Table

Ek comparison table banayein:

Model	Activation	Optimizer	Epochs	Performance
Model 1	ReLU	Adam	50	...
Model 2	Tanh	SGD	50	...
Model 3	ReLU	RMSprop	100	...

Aakhir mein explain karein ke kaunsa ANN configuration best perform karta hai aur kyun.

Important Note:

Sirf ek ANN model bana kar assignment complete nahi hoga. Aap ko different activation functions, optimizers aur ANN architectures experiment karni hain aur best performing model identify karna hai.

Goal: Model ki performance ko jitna possible ho improve karein, lekin overfitting se bhi bachna hai.

4. SUBMISSION CHECKLIST

Submit the following:

1. Jupyter Notebook / Google Colab file
2. Complete preprocessing code
3. ANN model code
4. Different experiments ke results
5. Final predictions
6. Model comparison

7. Short conclusion